

PROFESSIONAL CERTIFICATE IN MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE

Let's give everyone a couple of minutes to join...

Module 2

Fundamentals of Statistics and Distribution Functions

Toby Gardner

16 September 2025



Applied Stats & Distributions: A Data Scientist's Guide

When do we use distributions and statistics in real AI / ML projects?

- **Data Understanding:** involves Exploratory Data Analysis (“EDA”) - descriptive statistics and distributions
- **Data Preparation** (for model training) - where most Data Scientists spend their time - transformations, imputations, standardisation all use statistics → out of scope today
- **Model Selection** is dependent on the distribution of your dataset - out of scope for today
- ...basically, in every stage



Meet your Learning Facilitator!



Hi everyone, I'm Toby 🙋

- MBA (Masters in Business Admin.) from the University of California at Berkeley CA, USA.
- MAIDS (Masters of Applied Data Science) from the University of Michigan, MI, USA
- CFA Charterholder
- Co-founder Finch: financial SAAS using machine-learning to provide data-intelligence
- Prior to Finch, Toby was a Director at KPMG - where he was the top rated consultant in Australia

*This is what Midjourney says I should look like
(using my bio as a prompt)*

Office hours: Every other Tuesday

Always check Canvas for the most up-to-date information!

Tool to convert to your timezone: <https://www.worldtimebuddy.com/>

Office Hour Expectations

My objective: to get you to think like a real data scientist

- Understand at what level you will need to grasp AI ML concepts in practise ✓
- Understand when to *think* hard and move slow ✓
- Understand when to *move quickly* and break things ✓
- Foster a culture of collaboration and knowledge-sharing ✓
- Discuss ~~theoretical~~ real-world data and challenges ✓
- Discrepancies between course content ✗
- I can't open a notebook ✗
- Library not working ✗

Section 1

Exploratory Data Analysis ("EDA")

Quickstart on your Dataset - Descriptive Stats

When you're ready, move past `df.describe` and create your own functions

 `df_stats(df)`



=== Dataset Summary ===
The dataset consists of 5 columns x 150 rows

	dtype	vals	missing_pct	unique_vals	mean	mode	min	max	std	skew	kurtosis
sepal length (cm)	float64	150	0.0	35	5.84	5.0	4.3	7.9	0.83	0.3	-0.6
sepal width (cm)	float64	150	0.0	23	3.06	3.0	2.0	4.4	0.44	0.3	0.2
petal length (cm)	float64	150	0.0	43	3.76	1.4	1.0	6.9	1.77	-0.3	-1.4
petal width (cm)	float64	150	0.0	22	1.20	0.2	0.1	2.5	0.76	-0.1	-1.3
target	float64	150	0.0	3	1.00	0.0	0.0	2.0	0.82	0.0	-1.5

`df.describe()`



	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333	1.000000
std	0.828066	0.435866	1.765298	0.762238	0.819232
min	4.300000	2.000000	1.000000	0.100000	0.000000
25%	5.100000	2.800000	1.600000	0.300000	0.000000
50%	5.800000	3.000000	4.350000	1.300000	1.000000
75%	6.400000	3.300000	5.100000	1.800000	2.000000
max	7.900000	4.400000	6.900000	2.500000	2.000000

Quickstart on your Dataset - Descriptive Stats

Use statistics to get a quick 'pulse' on the data quality and work required

df_stats(df) #amazon best sellers 2025



= Dataset Summary ===
e dataset consists of 12 columns x 999 rows

	dtype	sample_val	vals	miss_pct	unique	mean	mode	min	max	std	skew	kurtosis
Unnamed: 0	int64	0	999	0.0	999	499.00	0.0	0.0	998.0	288.53	0.0	-1.2
rank	int64	1	999	0.0	100	50.45	1.0	1.0	100.0	28.85	0.0	-1.2
asin	object	B073VKKNN9	999	0.0	961	NaN	NaN	NaN	NaN	NaN	NaN	NaN
product_title	object	Kaspersky Premium - Total Security (Ultimate...	999	0.0	973	NaN	NaN	NaN	NaN	NaN	NaN	NaN
product_price	object	₹469.00	945	5.4	601	NaN	NaN	NaN	NaN	NaN	NaN	NaN
product_star_rating	float64	4.3	969	3.0	27	4.14	4.2	1.0	5.0	0.50	-1.7	7.8
product_num_ratings	float64	13324.0	969	3.0	485	1312.55	2.0	1.0	19189.0	3025.24	3.8	14.9
product_url	object	https://www.amazon.in/dp/B073VKKNN9	999	0.0	999	NaN	NaN	NaN	NaN	NaN	NaN	NaN
product_photo	object	https://images-eu.ssl-images-amazon.com/images...	999	0.0	941	NaN	NaN	NaN	NaN	NaN	NaN	NaN
rank_change_label	float64	NaN	0	100.0	0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
country	object	IN	999	0.0	10	NaN	NaN	NaN	NaN	NaN	NaN	NaN
page	int64	1	999	0.0	2	1.50	1.0	1.0	2.0	0.50	0.0	-2.0

Summary of what you just learnt

Pro tip: assume your dataset came from the bottom of a child's school bag, written in pencil, and rubbed in dirt

- Identified what fields you can probably drop for feature consideration e.g. asin, index, page
- Identified what data types you can turn into numbers for further analysis e.g. price → float, country → category
- Identified what feature engineering might be required e.g. USD conversion
- Identified what needs further investigation (see next section on Data Preparation)
 - Data duplication
 - Missing data



From descriptive stats to visualising distributions

Descriptive statistics are only half the story in EDA - don't make any decisions without visualising your distributions too. What distributions are these?

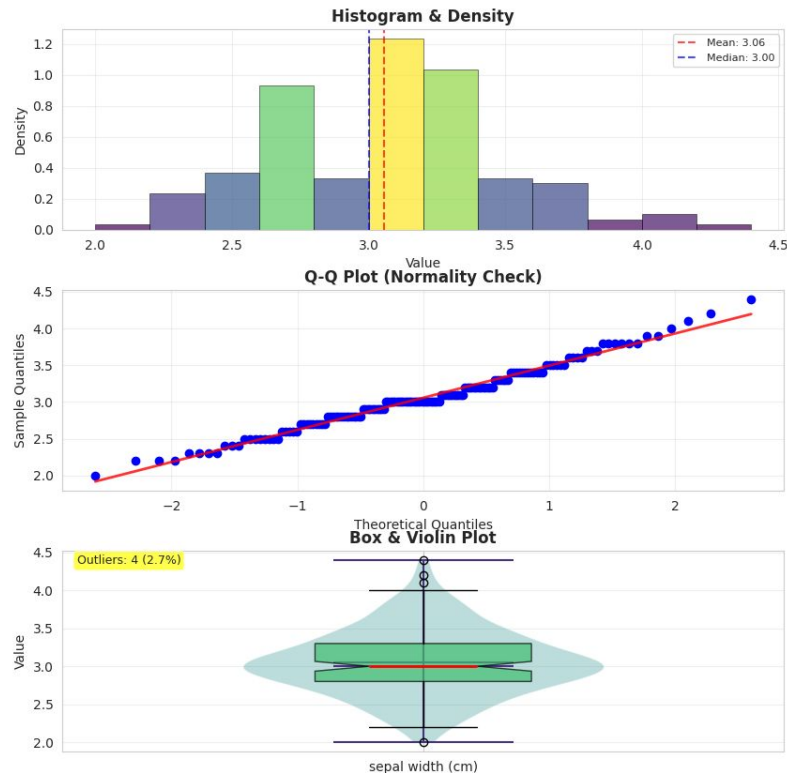
CENTRAL TENDENCY	
Mean:	3.0573
Median:	3.0000
Mode:	3.0000
Std Dev:	0.4359
Variance:	0.1900
CV:	0.1426
DISTRIBUTION SHAPE	
Skewness:	0.3158
Fairly symmetric	
Kurtosis:	0.1810
Normal tails	

CENTRAL TENDENCY	
Mean:	3.7580
Median:	4.3500
Mode:	1.4000
Std Dev:	1.7653
Variance:	3.1163
CV:	0.4697
DISTRIBUTION SHAPE	
Skewness:	-0.2721
Fairly symmetric	
Kurtosis:	-1.3955
Light tails (platykurtic)	

CENTRAL TENDENCY	
Mean:	1.0000
Median:	1.0000
Mode:	0.0000
Std Dev:	0.8192
Variance:	0.6711
CV:	0.8192
DISTRIBUTION SHAPE	
Skewness:	0.0000
Fairly symmetric	
Kurtosis:	-1.5000
Light tails (platykurtic)	

Normal Distribution

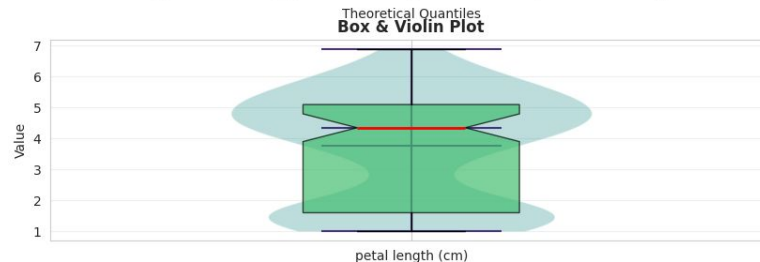
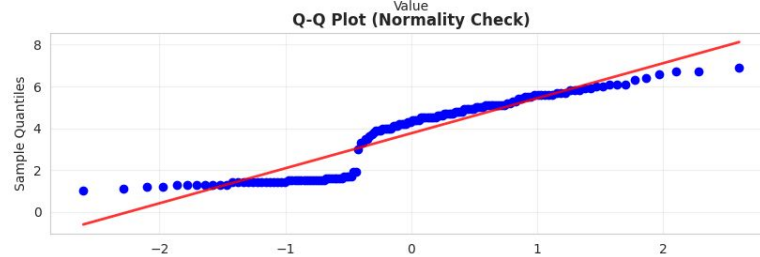
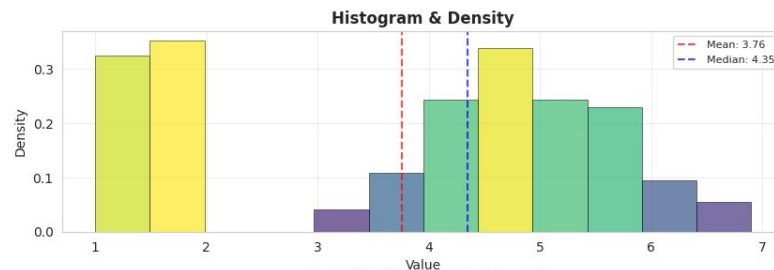
Distribution Analysis: sepal width (cm)



DESCRIPTIVE STATISTICS	
Count:	150
Missing:	0 (0.0%)
Unique:	23
Zeros:	0 (0.0%)
CENTRAL TENDENCY	
Mean:	3.0573
Median:	3.0000
Mode:	3.0000
Std Dev:	0.4359
Variance:	0.1900
CV:	0.1426
DISTRIBUTION SHAPE	
Skewness:	0.3158
Fairly symmetric	
Kurtosis:	0.1810
Normal tails	
RANGE & QUARTILES	
Min:	2.0000
Q1 (25%):	2.8000
Q2 (50%):	3.0000
Q3 (75%):	3.3000
Max:	4.4000
Range:	2.4000
IQR:	0.5000
OUTLIERS (IQR Method)	
Lower Bound:	2.0500
Upper Bound:	4.0500
Count:	4 (2.7%)
DISTRIBUTION TESTS	
Shapiro-Wilk p:	0.1012
✓ Normal	
Dip Test p:	0.0177
✗ Multimodal	
Estimated modes:	1
KS-Uniform p:	0.0000
CLASSIFICATION	
Multimodal (1 modes)	

Bimodal Distribution

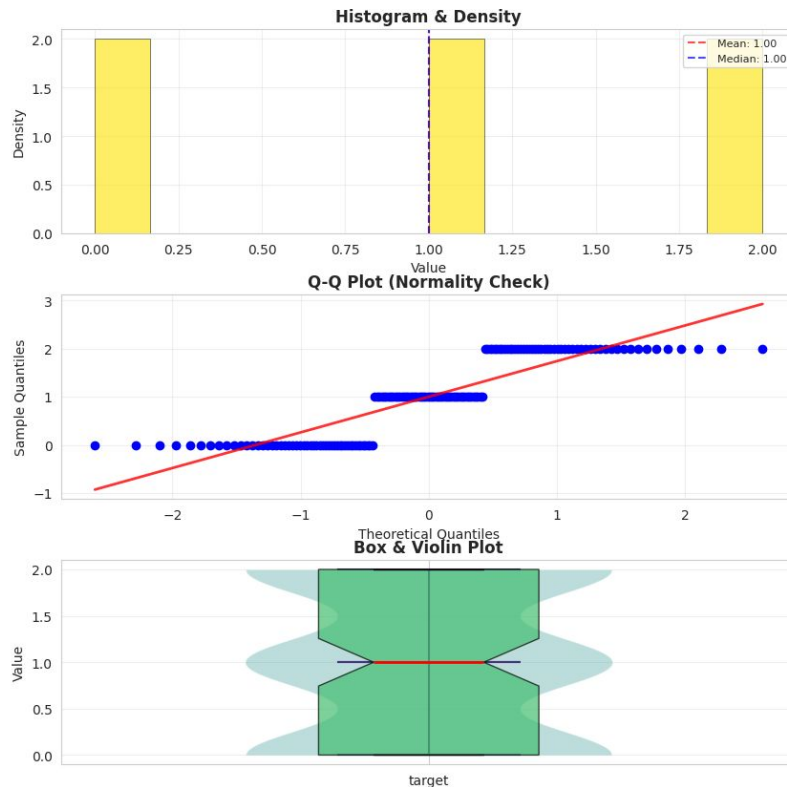
Distribution Analysis: petal length (cm)



DESCRIPTIVE STATISTICS	
Count:	150
Missing:	0 (0.0%)
Unique:	43
Zeros:	0 (0.0%)
CENTRAL TENDENCY	
Mean:	3.7580
Median:	4.3500
Mode:	1.4000
Std Dev:	1.7653
Variance:	3.1163
CV:	0.4697
DISTRIBUTION SHAPE	
Skewness:	-0.2721
Fairly symmetric	
Kurtosis:	-1.3955
Light tails (platykurtic)	
RANGE & QUARTILES	
Min:	1.0000
Q1 (25%):	1.6000
Q2 (50%):	4.3500
Q3 (75%):	5.1000
Max:	6.9000
Range:	5.9000
IQR:	3.5000
OUTLIERS (IQR Method)	
Lower Bound:	-3.6500
Upper Bound:	10.3500
Count:	0 (0.0%)
DISTRIBUTION TESTS	
Shapiro-Wilk p:	0.0000
x Not Normal	
Dip Test p:	0.0000
x Multimodal	
Estimated modes:	1
KS-Uniform p:	0.0000
CLASSIFICATION	
Multimodal (1 modes)	

Uniform Distribution

Distribution Analysis: target



DESCRIPTIVE STATISTICS	
Count:	150
Missing:	0 (0.0%)
Unique:	3
Zeros:	50 (33.3%)
CENTRAL TENDENCY	
Mean:	1.0000
Median:	1.0000
Mode:	0.0000
Std Dev:	0.8192
Variance:	0.6711
CV:	0.8192
DISTRIBUTION SHAPE	
Skewness:	0.0000
Fairly symmetric	
Kurtosis:	-1.5000
Light tails (platykurtic)	
RANGE & QUANTILES	
Min:	0.0000
Q1 (25%):	0.0000
Q2 (50%):	1.0000
Q3 (75%):	2.0000
Max:	2.0000
Recommended:	2.0000
Zero-inflated model	2.0000
OUTLIERS (IQR Method)	
Lower Bound:	-3.0000
Upper Bound:	5.0000
Count:	0 (0.0%)
DISTRIBUTION TESTS	
Shapiro-Wilk p:	0.0000
✗ Not Normal	
Data Type:	Discrete
Unique values:	3
✓ Discrete Uniform	
CLASSIFICATION	
Uniform	

Distribution Diagnostics Toolkit: Visual and Statistical Tests

Never trust summary statistics alone - distributions can remain hidden without proper diagnostics.

- Visual: Histograms, Q-Q plots, violin plots, CDFs
- Statistical tests: Shapiro-Wilk, Anderson-Darling, KS-test
- Skewness: > 1 or < -1 problematic
- Kurtosis: Excess > 3 indicates heavy tails
- Multi-modal detection: Hartigan's dip test



Summary of what you just learnt

Understanding why and what statistics / distributions mattered in EDA guides us to what steps to take in later AI / ML project phases

- Your Y (target) variable is uniformly distributed with 3 discrete values → **Model selection: classification model selection**
- An X feature looks normally distributed → **Data preparation: standardScaler**
- An X feature looks like bimodally distributed → few options
 - Classification models → Decision-Tree
 - Ask Qs to the data provider
 - Why is it like that?
 - Confirm you're not missing data

From EDA to data preparation: use of statistics still prevalent

You will often need to normalise or standardise your datasets before you input data into models for training

- Normalise data: scale feature values to within a predefined range, often between 0 and 1.
- Standardise: centres data around the mean (0) and scales it by the standard deviation (1)

Standardizing data ensures that each variable contributes equally to the training, preventing larger scale variables disproportionately influencing the result.

Variables measured in different units (e.g., kilograms, kilometers, etc.) should not influence the analysis based on their unit magnitude.

Pop Quiz: Before model training, we split our dataset into training, testing, and evaluation datasets. Would we want to do these transformations before or after the split?

Key Takeaways: Remember This

If you remember nothing else, remember that distributions are the foundation of reliable data science.

- Distributions reveal data quality issues before they become model failures
- Normal isn't always necessary, but know when it is
- Uniform distributions in real data are usually suspicious
- Transform thoughtfully - preserve information while meeting assumptions

...and use ChatGPT sparingly

- It is a terrible data scientist
- You will never understand data science if you don't understand 'what' matters and whether the answer looks 'right'

QUESTIONS?



Required Activities for Module 2

- Knowledge Check 2.1: Uniform Distribution Using Python
- Codio Assignment 2.1: Uniform Python
- Knowledge Check 2.2: Gaussian Distribution
- Knowledge Check 2.3: Multivariate Distributions
- Codio Assignment 2.2: Measures of Center and Spread
- Knowledge Check 2.4: Covariance
- Codio Assignment 2.3: Computing the Sample Covariance Matrix
- Knowledge Check 2.5: Correlation, Conditional Probabilities, and Independence
- Codio Assignment 2.4: Creating Histograms of Conditional Probabilities

Content review Module 2: Fundamentals of Statistics

- What's data science and why study statistics/probability?
- Probability vs Statistics
- Variables
- Descriptive statistics
- Correlation
- Probability distributions
- Central limit theorem